

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_3d035317-127\agent-aea9275be510a582b.jsonl`  
- SHA-256: `59e7407a3ae04d16a82efcb3d92fdb2aa6d16d8b492f1acf1065238837090e9`  
- Source modified: `2026-06-22T09:50:02+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_3d035317-127`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-22T09:46:23.308Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheelsutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file):
C:\Users\avidu\Projects\kheelsutra\docs\\LOCAL_APPLIANCE_ARCHITECTURE.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\tasks\w3dbzrpby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "docs/LOCAL_APPLIANCE_ARCHITECTURE.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. ``git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/`` or ``git -C ... show origin/master:backend/main.py``. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. ``backend/main.py::get_config``) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:26.431Z)

I'll start by reading the audit findings JSON and the target doc.

3. user (2026-06-22T09:46:28.971Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"
6      ],
7      "result": {
8      "perDoc": [
9      {
10     "doc": "NEXT_STEPS.md",
11     "overallStatus": "minor-staleness",
12     "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13     "findings": [
14     {
15     "claim": "PRs through #93 merged (master).",
16     "docLocation": "NEXT_STEPS.md:13",
17     "verdict": "stale",
18     "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19     "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20     "severity": "medium"
21     },
22     {
23     "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24     "docLocation": "NEXT_STEPS.md:5-14",
25     "verdict": "correct",
26     "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27     "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28     "severity": "low"
29     },
30     {
31     "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32     "docLocation": "NEXT_STEPS.md:7-9",
33     "verdict": "correct",
34     "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35     "suggestedFix": "",
36     "severity": "low"
37     },
38     ]
39     },
40     }
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38         {
39             "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40             "docLocation": "NEXT_STEPS.md:9-12",
41             "verdict": "correct",
42             "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43             "suggestedFix": "",
44             "severity": "low"
45         },
46         {
47             "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the DO droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48             "docLocation": "NEXT_STEPS.md:11-13",
49             "verdict": "correct",
50             "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51             "suggestedFix": "",
52             "severity": "low"
53         },
54         {
55             "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56             "docLocation": "NEXT_STEPS.md:13-14",
57             "verdict": "correct",
58             "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59             "suggestedFix": "",
60             "severity": "low"
61         },
62         {
63             "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64             "docLocation": "NEXT_STEPS.md:17-21",
65             "verdict": "correct",
66             "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67             "suggestedFix": "",
68             "severity": "low"
69         },
70         {
71             "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72             "docLocation": "NEXT_STEPS.md:32-41",
73             "verdict": "correct",
74             "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75             "suggestedFix": "",
76             "severity": "low"
77         },
78         {
79             "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80             "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",

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81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },
94     {
95         "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96         "docLocation": "NEXT_STEPS.md:44-45",
97         "verdict": "correct",
98         "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99         "suggestedFix": "",
100        "severity": "low"
101    },
102    {
103        "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104        "docLocation": "NEXT_STEPS.md:46-47",
105        "verdict": "correct",
106        "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107        "suggestedFix": "",
108        "severity": "low"
109    },
110    {
111        "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112        "docLocation": "NEXT_STEPS.md:65-73",
113        "verdict": "correct",
114        "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115        "suggestedFix": "",
116        "severity": "low"
117    },
118    {
119        "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120        "docLocation": "NEXT_STEPS.md:29-31",
121        "verdict": "unverifiable",
122        "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123        "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124        "severity": "low"
125    },
126    {

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127         "claim": "Control plane (control-plane/) at beta.",
128         "docLocation": "NEXT_STEPS.md:62",
129         "verdict": "correct",
130         "evidence": "control-plane/ directory does not exist in the repo (test -d
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131         "suggestedFix": "",
132         "severity": "low"
133     }
134 ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the
web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140     "findings": [
141     {
142         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143         "docLocation": "README.md:9",
144         "verdict": "stale",
145         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147         "severity": "high"
148     },
149     {
150         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151         "docLocation": "README.md:4",
152         "verdict": "stale",
153         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155         "severity": "medium"
156     },
157     {
158         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159         "docLocation": "README.md:17",
160         "verdict": "stale",
161         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162         "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163         "severity": "medium"
164     },
165     {
166         "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",

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167         "docLocation": "README.md:6",
168         "verdict": "correct",
169         "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170         "suggestedFix": "",
171         "severity": "low"
172     },
173     {
174         "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175         "docLocation": "README.md:7-8",
176         "verdict": "correct",
177         "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",
178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },

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215     {
216         "doc": "deploy/DEPLOY_ALPHA.md",
217         "overallStatus": "minor-staleness",
218         "summary": "Verified correct except a stale header",
219         "findings": [
220             {
221                 "claim": "header stale deploy live",
222                 "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223                 "verdict": "stale",
224                 "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225                 "suggestedFix": "set header to LIVE",
226                 "severity": "medium"
227             },
228             {
229                 "claim": "host list too narrow",
230                 "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231                 "verdict": "stale",
232                 "evidence": "engine.env.example adds loopback",
233                 "suggestedFix": "add localhost",
234                 "severity": "low"
235             }
236         ]
237     },
238     {
239         "doc": "deploy/droplet/README.md",
240         "overallStatus": "fresh",
241         "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entryptpoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242         "findings": [
243             {
244                 "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245                 "docLocation": "deploy/droplet/README.md:9-14",
246                 "verdict": "correct",
247                 "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248                 "suggestedFix": "",
249                 "severity": "low"
250             },
251             {
252                 "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253                 "docLocation": "deploy/droplet/README.md:53-54",
254                 "verdict": "correct",
255                 "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256                 "suggestedFix": "",
257                 "severity": "low"
258             },
259             {
260                 "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without

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it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",
271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import.meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,

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edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",
311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per $7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"

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337         },
338         {
339             "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341             "verdict": "stale",
342             "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343             "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344             "severity": "high"
345         },
346         {
347             "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",
348             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349             "verdict": "correct",
350             "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351             "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352             "severity": "low"
353         },
354         {
355             "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357             "verdict": "correct",
358             "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359             "suggestedFix": "",
360             "severity": "low"
361         },
362         {
363             "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365             "verdict": "correct",
366             "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367             "suggestedFix": "",
368             "severity": "low"
369         },
370         {
371             "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373             "verdict": "correct",
374             "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375             "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376             "severity": "low"
377         },
378         {

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379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited
:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",

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420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",
431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =

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[\"en\\\", \"kn\\\", \"hi\\\", \"es\\\", \"da\\\", \"id\\\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462     "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in §7 stay accurate as
historical record.",
463     "severity": "low"
464 },
465 {
466     "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467     "docLocation": "docs/DRY_RUN_CUJ.md:14",
468     "verdict": "correct",
469     "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly
as scoped.",
470     "suggestedFix": "",
471     "severity": "low"
472 },
473 {
474     "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475     "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476     "verdict": "correct",
477     "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
'local' and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == 'gs':
scheme = 'gcs'`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478     "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479     "severity": "low"
480 },
481 {
482     "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483     "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484     "verdict": "correct",
485     "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486     "suggestedFix": "",
487     "severity": "low"
488 },
489 {
490     "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491     "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492     "verdict": "correct",
493     "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494     "suggestedFix": "",
495     "severity": "low"
496 },
497 {

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498         "claim": "DoD §6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",
507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose §9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "§3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; §4.6 Reuse map lists it under 'Net-new'; §9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of §4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (?/?), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email` `owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable

```

```

(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[#66]', T12 'SPA
#56', alpha-deploy '[#64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ?: 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",

```

```

570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 §9 records the live in-UI gs://?reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git
log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "§3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

4. user (2026-06-22T09:46:29.413Z)

```

1      # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3      > **Status:** `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
bridge) . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this
repo; **consumes** the engine's decoupled `StorageBackend` + `python -m backend.worker` seams
and the FastAPI loop; **extends** the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha
screens. Engine-side changes here are **proposals to coordinate** with the active
decoupled/worker sessions ? not written unilaterally.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     **KhelSutra Studio** turns the engine from a CLI into a **single-box, UI-powered rally
indexer + reel generator**: the khelsutra `web/` React SPA, served behind an **edge auth gate**,
drives the **existing** engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on
**one machine**, with `StorageBackend=local` for the MVP. It is the **single-box deployment
profile of the just-shipped decoupled architecture** ? storage=local + compute=local is the
degenerate collapse of the same storage/worker split.
13
14     The **single most important caveat:** the engine has **zero app-layer auth and CORS
`***` `[verified backend/main.py:42]`, so safety is structural **only** because the engine stays

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bound to `127.0.0.1:8000` `[verified backend/main.py:638]`, a firewall blocks inbound `8000`,
and a reverse proxy/tunnel is the sole authenticated ingress ? the gate, not the app, is the
tenancy boundary.
15
16     Two honest hardware truths: (1) the current droplet is CPU-only ? real detection
there is Gemini (cloud) or stub (CPU mock); real local-GPU detection needs a GPU
host (native WASB, in-flight). (2) The owner priority is to stand up a GPU host ASAP ?
provider-agnostic (DO GPU droplet / GCE GPU VM / Cloud Run+GPU) ? but that runs in parallel
with the UI MVP, which ships on Gemini today.
17
18     ## 1. Problem & goal
19     The decoupled engine just shipped a `StorageBackend` ABC + a `python -m backend.worker
{infer|train}` dispatcher `[verified backend/storage/, backend/worker/__main__.py]`, and the
FastAPI loop already covers `process ? jobs ? query ? compile ? highlights` `[verified
backend/main.py]`. What's missing is a console that lets a non-CLI operator run the whole
loop on a single box, honestly, behind a gate.
20
21     Headline CUJ (owner's words): a hosted website where I pick an input ? a
Google Drive file, a bucket file (`s3://`/`gs://`), or a local upload ? it lands in the
bucket (DO Spaces / GCS) or locally ? I generate highlights on a GPU-enabled host
running the real e2e ? I pick rallies, watch, and download ? completely from the UI.
22
23     Outcome ("good"): an operator signs in, picks a source, watches it index, curates
rallies, downloads a reel ? on the live CPU droplet today (Gemini/stub), and on a GPU host
later (native WASB) with no UI fork.
24
25     Read to get current: `PER_RALLY_SELECTION.md` ($3/$4/$10 contract anchors), engine
`HOSTING_PLAN.md` (gating), `DECOUPLED_COMPUTE/HANDOFF.md` (what's built),
`VIDEO_LOCALITY_MODEL.md` (privacy/retention).
26
27     ## 2. Decisions locked
28     | # | Decision | Source / date | Implication |
29     |---|---|---|---|
30     | D1 | Reverse-proxied single origin; engine stays `127.0.0.1:8000` | `[verified
main.py:638]`; 2026-06-20 | CORS `` is moot from the browser ? the proxy is the boundary |
31     | D2 | Edge auth, zero engine auth code | `HOSTING_PLAN.md`; 2026-06-20 | Caddy
basicauth OR cloudflared + Cloudflare Access |
32     | D3 | MVP drives the in-process FastAPI worker loop, `storage=local` | `[verified
job_worker.py:37, local.py:33]`; 2026-06-20 | bucket worker = reserved scale path, out of MVP |
33     | D4 | Embed RallyPicker from `PER_RALLY_SELECTION.md` verbatim | PR [#11];
2026-06-20 | the console is a superset; do not redesign the picker |
34     | D5 | Console lives in the khelsutra web React scaffold | `web/README.md`;
2026-06-20 | Cloudflare-Pages-style root, not engine static |
35     | D6 | Compute placement is a setting, rendered from `GET /api/capabilities` |
`[design]`; 2026-06-20 | UI offers ONLY what the box can honestly do |
36     | D7 | Ingest is a source menu (local upload · bucket URI · Drive) normalizing to
one video object | `[verified backend/storage/ref.py parse_uri]`; 2026-06-20 | Drive is a
stub today ? scoped honestly ($4.2) |
37     | D8 | GPU host is provider-agnostic (DO droplet / GCE VM / Cloud Run+GPU); storage
stays the bucket | Owner 2026-06-20 | compute and storage are separate planes; no provider lock-
in |
38     | D9 | Observability is first-class ? structured logs on every new path + a DoD item
| Owner 2026-06-20 | see $5; verified during CUJ replay |
39     | D10 | Delivery = a self-hosted webserver + UI the user spins up (NOT a desktop
app); the user points at local storage + GPU/CPU OR cloud locations and the UI drives the
tool | Owner 2026-06-20 | supersedes `DESKTOP_APP_PLAN.md`; one artifact (this appliance)
runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings |
40
41     ## 3. Foundation ? what the engine gives us today `[verified this session unless
noted]`
42     - FastAPI loop: `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ?
`POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}`
(FileResponse) `[main.py]`. Chunked upload exists ? `POST /api/upload/init` · `PUT
/api/upload/{id}/part/{i}` · `POST /api/upload/{id}/complete` · `DELETE /api/upload/{id}`
`[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` never calls them (it only

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hits `/api/process` + `/api/compile` `[verified app.js:88,373]`. So browser upload is client-
wiring only, no engine change.
43   - In-process worker: one module-global `ThreadPoolExecutor(max_workers=1)`
  `[verified job_worker.py:37]` ? strictly one job at a time; the DB `jobs` table is the
  clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
  separate knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
  to serialize) ? it is not already 1.
44   - Storage: `storage.backend` default = "local" `[verified models.py:298]`;
  `LocalStorage.fetch_to_local` is identity `[verified local.py:33]`, `put` no-ops on
  `src==dst` ? so on a single box the worker's pull?pipeline?push collapses to read-local/write-
  local. URI scheme `parse_uri` `[verified ref.py]`: bare/`local://` ? local; `s3://`
  (AWS/R2/DO Spaces/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? follow-up stub
  (emulated only). Credentials never in config ? boto3 default chain / `~/aws` `[verified
  s3.py]`.
45   - Compute seam: `indexing.detector_impl` ? {`auto`=WSL, `native`=Linux-native,
  `stub`=CPU mock}; default "auto" `[verified models.py:129]`, resolved in
  `trajectory_hybrid._resolve_runner` `[verified :76-85]`. Native WASB is in-flight on branch
  `feat/native-wasb-runner`: native_wasb_runner.py exists `[verified]`;
  fusion_hybrid._build_native_runner returns `NativeWasbRunner` exists `[verified
  fusion_hybrid.py:86]` while trajectory_hybrid._build_native_runner still raises not
  verified `[verified :57-63]`. Acceptance gate = owner byte-parity (WSL CSV == native CSV) on a real GPU box ?
  not exercisable on the droplet or in the offline suite. Treat native as not production-
  verified.
46   - Worker CLI: `python -m backend.worker {infer|train}` pulls a `--video-uri` ?
  pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified
  worker/__main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
  `--set indexing.skip_ai_handoff=true` + `storage.s3`.
47   - Gating: cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
  email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
  /api/health` now EXISTS (engine M1, [#268]) ? {`status`, `sport`, `default_segmenter`}
  `[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.
48   - Droplet reality: `168.144.126.176`, CPU-only, NO GPU, region `blr1`; co-hosts
  the marketing site (`/srv/khelsutra-site`, node `:8787`) and the engine (`~/rally`); ran the
  full train+infer e2e against DO Spaces with the GPU mocked `[SETUP_NEW_MACHINE.md,
  DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
  there.
49   - Absent (net-new): `GET /api/capabilities`, `GET /api/videos` (only single-row
  `get_video(id)`), `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any
  Dockerfile, any per-video cost ledger `[verified ? grep returns none]`.
50
51   ## 4. Design / architecture
52
53   ### 4.1 Topology & request path
54   ONE box: edge gate (cloudflared+Access or Caddy basicauth) ? reverse proxy /
  static host serving the `web/` SPA bundle and proxying `/api/*` ? engine FastAPI
  `127.0.0.1:8000` ? LocalStorage ? detector seam (Gemini/stub today; native WASB on a
  GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
  site stays a separate systemd unit on a separate subdomain.
55
56   ...
57   PUBLIC INTERNET
58   ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
59   ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
  basicauth
60   ?
61   ?????????????????????????????????????????????????????????????????????????????????
62   ? ONE BOX firewall BLOCKS inbound 8000 + 9000/9001/9023 ?
63   ? (2) reverse proxy / static host (Caddy/cloudflared) ?
64   ? /, /assets/* ? web/ React SPA /api/*, /api/highlights/* ? :8000 ?
65   ? ? ONE ORIGIN to the browser ? engine CORS '*' is irrelevant ?
66   ? ? ?
67   ? (3) ENGINE FastAPI (host=127.0.0.1:8000, HARD-bound [main.py:638]) ?
68   ? control: jobs table + ThreadPoolExecutor(max_workers=1) drain ?
69   ? data: uploads.py chunked upload . /api/compile . /api/highlights ?
70   ? ? ?

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71      ? (4) STORAGE = LocalStorage (default)  fetch=identity · put=no-op(src==dst) ?
72      ?                                     ?
73      ? (5) COMPUTE seam: gemini (cloud, key) · stub (CPU mock) [droplet]          ?
74      ?                                     native WASB (GPU host) · auto (owner Win+WSL) ?
75      ?????????????????????????????????????????????????????????????????????
76      RESERVED SCALE PATH (out of MVP): console "compute target" selector ? storage=s3 (DO
Spaces)
77      ? SEPARATE GPU host runs `python -m backend.worker infer --video-uri s3:///` ?
outputs/<vid>/ back
78      (needs a NET-NEW orchestration endpoint reflecting outputs into the jobs/poll
contract)
79      ...
80
81      ### 4.2 Ingest sources (the headline) ? one menu, one normalized video object
82      All sources normalize to ***a video the pipeline can read*** ? a local path (MVP) or a
`StorageRef` (scale):
83
84      | Source | Path today | Status |
85      |---|---|---|
86      | **Local upload** | SPA chunks the file ? `max_part_mb` via the existing
`/api/upload/init\|part\|complete` `[verified uploads.py]` ? server path ? `POST /api/process`.
(Scale: also `StorageBackend.put` to `s3:///`/`gcs:///`.) | **wire client only** (endpoints exist)
|
87      | **Bucket URI** (`s3:///`, `gs:///`) | **POST /api/process` accepts the bucket URI
directly** (resolve ? fetch to scratch ? same pipeline ? API DB) `[verified main.py:159,274,283
? corrected 2026-06-21; was "net-new"]`; the worker CLI also fetches `--video-uri`. So the
**engine endpoint is NOT net-new** ? only the SPA call that POSTs the URI + polls `/api/jobs`
is. *(Default single-user mode; HOSTED mode w/ `allowed_video_root` correctly rejects a bucket
URI 400.)* | **engine: DONE (verified); SPA ingest call: design (= B-P2)** |
88      | **Google Drive** (`gdrive:///`) | `parse_uri` recognizes it, but the gdrive backend is
an **emulated stub** `[verified ref.py, DECOUPLED_COMPUTE research]` | **stub ? roadmap;** MVP
requires a Drive?bucket pre-step; do **not** present Drive as live until the backend is real |
89
90      ### 4.3 Compute placement matrix (one UI, surfaced from `/api/capabilities`)
91      | Placement | Runs | Needs | Real/mock | Honest caveat |
92      |---|---|---|---|---|
93      | **CPU droplet ? gemini** (MVP target) | `segmenter=gemini`, in-proc worker |
`GEMINI_API_KEY`; network; **no GPU** | **REAL** | Cloud brain: uploads ?1280px chunks to Google
? **must be in consent copy**; **no per-video cost ceiling** today |
94      | **CPU droplet ? stub** | trajectory hybrid + `detector_impl=stub`+`skip_ai_handoff` |
nothing | **MOCK** | not real tracking ? UI must label "stub (CPU mock ? not real detection)";
demo/regression only |
95      | **CPU droplet ? motion** | `segmenter=motion` | nothing | REAL, low-quality | only no-
key/no-GPU real segmenter; never surface `server/receiver/events` (fabricated) |
96      | **GPU host ? native WASB** (in-flight) | `wasb`+`detector_impl=native` ?
`NativeWasbRunner` (in-proc torch) | a **GPU host ? DO GPU droplet / GCE GPU VM / Cloud
Run+GPU**; `WASB_WEIGHTS` | **REAL**, on-device | partially wired (fusion builds it, trajectory
refuses); **owner byte-parity gate**; **Cloud Run/GCE need a containerized worker ? no
Dockerfile exists, and WASB's torch-1.11 conda env makes the image non-trivial** |
97      | **Owner Win+WSL ? auto** | `wasb`+`detector_impl=auto` (WSL2+conda) | Windows+WSL2+GPU
| REAL (today) | Windows-only transport; the working real-GPU path for owner verification |
98      | **Remote GPU worker via bucket** (reserved) | console CPU + `storage=s3`; separate GPU
host runs `backend.worker infer` | bucket creds; a GPU host; a **net-new orchestration
endpoint** | REAL (remote) | FastAPI loop and worker CLI are **separate** today; "remote GPU as
a setting" is **design** ? selector shown **disabled** in MVP |
99
100     **Provider note:** **Cloud Run + GPU** (scale-to-zero, pay-per-job) is the strongest fit
for the engine's own **per-video-billing** economics (`DECOUPLED_COMPUTE` #246) ? gated on the
**containerization feasibility check** above.
101
102     ### 4.4 Auth / exposure (non-negotiable invariant)
103     No engine auth, by design; the gate is external. **Invariant (all three must hold):**
engine bound to `127.0.0.1` + firewall closes `8000` + edge gate up. Go-live checklist
`[HOSTING_PLAN.md:102-108]`: Access/basicauth ON; **rotate the chat-shared `GEMINI_API_KEY`**;
set `security.allowed_video_root`; rate-limit; per-user quota.

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104
105     ### 4.5 UI operator console (web/ SPA ? superset of RallyPicker)
106     - **S1 Ingest & Process** ? the source menu ($4.2) + a segmenter picker from
    `/api/capabilities` ? `POST /api/process`.
107     - **S2 Progress** ? 3s poll of the free-text stage
    (`hashing/validating/segmenting/finalizing`), a one job at a time notice, and the restart-
    failure terminal state.
108     - **S3 Rally pick ? reel** ? embed RallyPicker verbatim (`PER_RALLY_SELECTION.md`):
    honor `min_shot_count`; never render motion `server/receiver/events` or a fake confidence meter.
    Per-rally is ONE screen of the console.
109     - **S4 Reels / History** ? needs `GET /api/videos` + `GET /api/reels/{video_id}` for
    reload-survival.
110     - **S5 Settings** ? render `config` + `capabilities`; the compute/storage target
    selector shown DISABLED ("single-box: local") until the bucket-worker bridge lands.
111
112     ### 4.6 Reuse map
113     Reused (no re-plumb): the full `/api` loop; the in-process worker; the chunked
    upload endpoints (un-called); LocalStorage default; the detector seam incl. the landing
    `NativeWasbRunner`; the RallyPicker design + anchors; the `web/` scaffold; the
    `HOSTING_PLAN` gate; the `SETUP_NEW_MACHINE` runbook. Net-new: Caddyfile + systemd units;
    `GET /api/capabilities`; `GET /api/videos`; `GET /api/reels/{video_id}`; `GET /api/health`; the
    SPA console screens; ingest-from-bucket wiring; a retention sweep; (reserved) the bucket-worker
    bridge.
114
115     ## 5. Observability ? logging & monitoring `[required, first-class]`
116     Launch-readiness directive (owner 2026-06-20): logs and monitoring taken seriously,
    with a full log-analysis pass across every new code path. The engine already uses a
    centralized Python logger `[verified main.py:12-17]` ? carry that discipline into every new
    path.
117
118     - Correlation ID end-to-end. Thread the existing `job_id` (from `/api/process`)
    plus an `upload_id`/`request_id` through ingest ? process ? jobs ? compile ? highlights, and
    into the worker for the bucket path, so one CUJ is traceable across UI ? FastAPI ? worker ?
    GPU host. Log it on every line.
119     - Structured logs. New endpoints/screens emit structured (JSON-able) logs at entry
    / result / error ? not just "it ran". The SPA logs client-side errors + failed fetches with
    the same correlation id.
120     - Loud failures, never silent drops. Carry the engine's ethos (#240 "surface failed
    chunks loudly") + RallyPicker's `min_shot_count`-warning rule: a partial/empty reel must
    never look complete; every dropped clip/chunk is logged + surfaced in the UI.
121     - Health + metrics. Ship the missing `GET /api/health` `[HOSTING_PLAN.md:42]` and a
    thin metrics surface (jobs queued/running/failed, last error, disk free, per-video cost once a
    ledger exists) for the engine-mode banner + alerting.
122     - Aggregation for transient hosts. A scale-to-zero Cloud Run / GPU host is
    ephemeral ? logs must ship to a durable sink (the bucket or a log service), not die with the
    instance.
123     - Log-analysis pass before merge (DoD item). Every PR that adds a code path includes
    a check that the path logs meaningfully, and the logs are inspected during CUJ live-replay
    ($8 Layers 173). Ties [[feedback-launch-quality-bar]].
124
125     ## 6. Risk table
126     | Risk | Today | Mitigation |
127     |---|---|---|
128     | No app auth + CORS `` `[main.py:42]` | open if any of {localhost-bind, firewall,
    gate} slips | enforce all three; LIVE posture test that raw `:8000` is refused from outside
    |
129     | Test ports on shared droplet | MinIO `9000/9001` + GCS emu `9023` | firewall closes
    them |
130     | Retention | sweep tool now exists ([engine
    #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299), dry-run-first, uploads
    >7d, reels durable) but not yet auto-scheduled ([engine
    #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)) | run the sweep
    `--apply` before/with any non-owner exposure; auto-schedule per
    [#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301); keep-proxy-only
    still an option |

```

```

131 | Surprise Gemini/GPU bill | **no cost ceiling/ledger** | per-video cap;
`ai_max_workers` (default 5) tunable; `max_workers=1` job serialization |
132 | Native WASB unverified | partially wired; owner-parity-gated | treat as not-
production-verified until owner byte-parity on a real GPU box |
133 | New engine endpoints | `capabilities`/`videos`/`reels`/`health` don't exist |
coordinate engine PRs with active sessions; **branch from `origin/master`, re-verify HEAD,
explicit `git add`** |
134 | Drive ingest overstated | `gdrive://` is a stub | don't present Drive as live; require
Drive?bucket pre-step until real |
135 | No Dockerfile | runbook-only reproducibility | feasibility-check a worker image before
promising Cloud Run/GCE |
136 | Shared-checkout collisions | active worker sessions share the engine tree | the
branch-safety protocol on every engine-side PR |
137 | **Provisioning-credential leak** | a DO API token / cloud creds could leak if pasted
in chat or committed (cf. the **already-rotated Spaces key** exposed in chat,
`DECOUPLED_COMPUTE/HANDOFF.md`) | authenticate the CLI **on-box** (`doctl auth init` interactive
/ `gcloud` already authed) so the secret never enters chat or the repo; least-privilege token,
revoke after; creds 0600, never committed; **creating paid GPU infra is a confirm-first action
every time** |
138
139 ## 7. Honest limits ? what this does NOT do
140 Does **not** add multi-tenant identity/RLS (single-tenant alpha; uploads/outputs
global). Does **not** run real WASB on the CPU droplet (Gemini or stub only). Does **not** make
the remote-GPU bucket path work (design; selector disabled). Does **not** provide in-UI source
playback, job cancel, or a true progress bar (coarse free-text stage, poll-only). Does **not**
containerize (runbook; no Dockerfile `[verified]`). Does **not** verify native WASB (owner byte-
parity on a real GPU box, not this offline suite). A green SPA + contract tests prove the
**console plumbing**, not engine rally-detection quality (a separate track).
141
142 ## 8. CUJs & live-test plan `[required]`
143 **CUJs:** **CUJ-1** ingest (Drive/bucket/local) ? index ? reel ? download (core);
**CUJ-2** re-open a past match after reload; **CUJ-3** pick-by-query (deterministic filters +
honesty-hiding); **CUJ-4** settings/compute placement (truthful capabilities).
144
145 **Live-test plan ? 5 layers, every CUJ replayed LIVE throughout development:**
146 - **L1 ? Mock engine (hermetic, fast, default + CI).** A tiny FastAPI/MSW implementing
the **exact** contract (upload; `process` 202; `jobs` state machine incl. restart-failure;
`query` with a motion fixture for honesty-hiding; `config`; `capabilities` per box profile ?
CPU-no-key / CPU-with-key / GPU; `compile`; `highlights` tiny mp4; the new `videos`/`reels`).
The SPA drives **all 4 CUJs in <1s**, **replayed via the Claude Preview/browser tools** (start
dev server ? fill the ingest picker ? click through process/poll/RallyPicker ? assert the reel
link via snapshot/network). Catches reload-survival (CUJ-2), the `play_segments` field-name bug
(CUJ-3), capability honesty (CUJ-4).
147 - **L2 ? Engine-contract checks (real FastAPI, GPU-free).** Boot `python -m backend.main
--server` with `storage=local` + `detector_impl=stub` + `skip_ai_handoff` and assert the **mock
and the real engine agree shape-for-shape** (202 keys, jobs enum, `result.play_segments` schema,
compile `{status,filepath}`, highlights content-type). Every new endpoint is born with a mock
handler **and** a real-engine contract test **in the same PR**. Runs in CI.
148 - **L3 ? Real local e2e (the MVP substrate).** Through the gate on the droplet: CUJ-1
with `stub` (key-free) and `gemini` (real, short clip to bound cost); the **exposure-safety
posture test** (raw `:8000`/`:900x` refused from outside) as an assertion, not an assumption;
reuse the `parity.yml` dual-host pattern.
149 - **L4 ? CI regression (green gates merge).** `npm test` + a stub-engine contract job +
headless browser-replay CUJ scripts; one small PR per tracker row off `origin/master`, **no
stacking**; coverage at/above the floor; **logs inspected during replay** (§5).
150 - **L5 ? Owner hardware e2e (handed off).** Native WASB **byte-parity** on a real GPU
box; real-4K **golden-corpus** runs (Gemini + native); Modern-Standby/`cv2`-vs-`ffmpeg`
behavior; a multi-GB browser upload. Reported back into §9, stated plainly as **not verified**
in the offline suite.
151
152 ## 9. Deliverables & progress tracker ? **source of truth**
153 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR
ships its tests + meaningful logs (§5).
154

```

```

155 | ID | Deliverable | Depends on | Gated? | Status | PR |
156 |----|-----|-----|-----|-----|----|
157 | T0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | owner | ? |
(this PR) |
158 | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |
159 | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160 | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
161 | T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66)) ** ? file picker ? chunked `/api/upload/`
? server path ? `/api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
162 | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163 | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164 | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165 | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166 | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167 | T10 | Observability pass: correlation id end-to-end + structured logs + metrics (S5) |
T4 | no | ? | ? |
168 | T11 | Retention sweep for `output/`+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-` + regenerable frames/staged/proxies; **reels/DB/CSVs
 durable**;  `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
169 | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs
(M2/[#280]) *** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` §9) | T4 | ? | ? | engine #280 + SPA #56 |
170 | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner
byte-parity (P4) | branch | ? owner | ? | ? |
171 | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
172
173 **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does **not** block on it (ships on Gemini).
174
175 *** Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).**
The alpha ships the **split** topology (SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel + CF Access over both ? the engine `HOSTING_PLAN.md` §2), **not** the single-
origin Caddy shell of **T2** (kept as the documented alternative ? §0/D1). The **deploy kit +
runbook** ([`deploy/DEPLOY_ALPHA.md`](../deploy/DEPLOY_ALPHA.md) +
`deploy/cloudflared.config.example.yml`) is shipped (? templates+runbook); the actual stand-up
(tunnel + Pages + CF Access + key rotation) is the **owner** step. The **T3**
raw-`:8000`-refused posture test is documented as the deploy kit's §6 verification. Enabled by:
engine exposure-safety boot posture (engine #294) + SPA `VITE_API_BASE` ([#64]).
176
177 ## 10. Open questions ? owner / external
178 **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:**
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.
179 **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming

```

```

endpoint; per-user scoping timing; containerization vs runbook.
180
181     ## 11. Definition of done
182     - [ ] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via
the ingest menu.
183     - [ ] Raw `:8000`/`:900x` unreachable from outside (**tested**, not assumed).
184     - [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped
honestly (not faked).
185     - [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
186     - [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via
`/api/videos` + `/api/reels`.
187     - [ ] **Observability:** correlation id threaded end-to-end; structured logs at every
new path; loud failures; logs inspected during CUJ replay.
188     - [ ] Retention sweep in place (**tool shipped** [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-schedule**
[engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301));
`GEMINI_API_KEY` rotated; `allowed_video_root` set.
189     - [ ] CI green (L2 contract + L4 regression); coverage ? floor; $9 tracker all
??-resolved.
190
191     ## 12. References
192     **Engine code `[verified this session]`:**
`backend/main.py:12-17,42,52,307-328,331-340,342-396,638`;
`backend/api/uploads.py:64,89,123,147`; `backend/api/job_worker.py:37`;
`backend/api/static/app.js:88,344,373`;
`backend/storage/{base.py,local.py:33,ref.py,__init__.py:43}`;
`backend/config/models.py:129,183,298`;
`backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86,
gemini.py:470-485, motion.py:191-198}`; `backend/pipeline/detectors/{native_wasb_runner.py,
stub_runner.py}`; `backend/worker/__main__.py:88,98,149`; `config.json` (`default_segmenter:
gemini`). **Engine docs:** `HOSTING_PLAN.md:42,102-108`; `VIDEO_LOCALITY_MODEL.md`;
`DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`;
`DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`;
`PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`,
`PORTABILITY.md`, `web/README.md`.
193
194     ---
195     ### Changelog
196     - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs
(FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin
spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor
re-verified against source before writing (catching the native-WASB partial-wiring +
`ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local),
multi-provider GPU, and first-class Observability directives.
197

```

5. assistant (2026-06-22T09:46:33.119Z)

Let me read the rest of the audit findings for this doc and verify the engine claims. Let me get the remaining findings for this specific doc.

6. user (2026-06-22T09:46:35.334Z)

```

591         },
592         {
593             "claim": "D10 / §12: this appliance design 'supersedes
DESKTOP_APP_PLAN.md'.",
594             "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:39,192",
595             "verdict": "unverifiable",
596             "evidence": "DESKTOP_APP_PLAN.md is an engine-repo doc, not present in the
khelsutra docs/ tree (git ls-files khelsutra has no DESKTOP_APP_PLAN.md). Cannot confirm its
current supersession status from this repo.",
597             "suggestedFix": "",
598             "severity": "low"
599         }

```

```

600     ]
601   },
602   {
603     "doc": "docs/MULTILINGUAL_EXPANSION.md",
604     "overallStatus": "minor-staleness",
605     "summary": "An accurate, still-live tracker: its core claims hold up ? Phase A
(es/da/id) shipped exactly as described (#67), the locale catalogs live at the stated repo-root
/locales path, and Phases B (Thai) / C (CJK) genuinely have not started (no th/zh/ja/ko catalogs
or fonts in the tree). The staleness is minor and concentrated in two places: (1) EX3 and the
repeated \"SPA-only until FRIENDLY B3\" framing are now outdated ? B3 (#70) and site es/da/id
parity (#87) both shipped, so the marketing site is no longer SPA-only for the Phase-A trio; (2)
a broken $10 reference to a non-existent docs/UI_PROPOSAL.md, plus small line-number/test-count
drift. Not an archive candidate: Phases B/C are real open work.",
606     "findings": [
607       {
608         "claim": "Phase A (es/da/id) shipped via #67; E0/EA1/EA2/EA3 ? (E0 via
owner-merged #58).",
609         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,75-78",
610         "verdict": "correct",
611         "evidence": "git log: a2292ef 'feat(web): MULTILINGUAL Phase A ?
Spanish/Danish/Indonesian SPA catalogs (#67)'; a20002f 'docs: propose Multilingual Expansion ?
(#58)'. locales/{es,da,id}/common.json all tracked.",
612         "suggestedFix": "",
613         "severity": "low"
614       },
615       {
616         "claim": "Catalogs live at repo-root
/locales/{en,kn,hi,es,da,id}/common.json (hoisted by FRIENDLY B3, D-CATALOG-HOME).",
617         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:51,108",
618         "verdict": "correct",
619         "evidence": "git ls-files locales/ returns exactly en,kn,hi,es,da,id
common.json; web/src/i18n/index.ts:16-21 imports them from
'../../../locales/<loc>/common.json'.",
620         "suggestedFix": "",
621         "severity": "low"
622       },
623       {
624         "claim": "Phases B (Thai) and C (zh-Hans/ja/ko) are owner-gated and not
started (EB*, EC* all ?).",
625         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,79-84",
626         "verdict": "correct",
627         "evidence": "git ls-files | grep 'locales/(th|zh|ja|ko)' returns nothing
(exit 1); grep for noto-sans-(thai|sc|jp|kr) in web/site package.json returns nothing (exit
1).",
628         "suggestedFix": "",
629         "severity": "low"
630       },
631       {
632         "claim": "EX3 ('site/ marketing parity for the 7') is ? and 'Blocked on B3';
and $1/$4/$6 say site localization 'depends on FRIENDLY B3 landing first ? until then this is
SPA-only.'",
633         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
634         "verdict": "stale",
635         "evidence": "FRIENDLY B3 shipped: git log 1b1fa7b 'feat(web+site): B3 ?
trilingual marketing site via shared t()/data-i18n shim (#70)'. Site es/da/id parity also
shipped: cf59fbf '#87'. git ls-files shows site/public/locales/{es,da,id}/common.json present.
The dependency is satisfied and the Phase-A trio is no longer SPA-only.",
636         "suggestedFix": "Update EX3 to reflect that FRIENDLY B3 landed (#70) and the
Phase-A es/da/id site parity shipped (#87) ? mark the es/da/id portion ? (site has those
catalogs) with only th/zh/ja/ko site parity remaining; drop the 'until then this is SPA-only'
caveat for es/da/id in $1/$4/$6.",
637         "severity": "medium"
638       },
639       {
640         "claim": "$10 References cites internal 'docs/UI_PROPOSAL.md'.",

```

```

641         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
642         "verdict": "stale",
643         "evidence": "git ls-files | grep -i UI_PROPOSAL returns nothing (exit 1) ?
no such file exists anywhere in the repo.",
644         "suggestedFix": "Remove the docs/UI_PROPOSAL.md reference or repoint it to
the correct doc (the engine repo has UI docs; this repo does not contain UI_PROPOSAL.md).",
645         "severity": "low"
646     },
647     {
648         "claim": "Changelog: '168 web tests green' (as of #67, 2026-06-21).",
649         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:114",
650         "verdict": "correct",
651         "evidence": "That was the count at the #67 changelog timestamp and is framed
as a point-in-time changelog note. The live suite now reports 223 passed (24 files) via `npx
vitest run` ? higher because of later PRs (#68/#70/#73/#80/#88 etc.), so the historical 168
figure is not wrong, just superseded.",
652         "suggestedFix": "Optional: the changelog is dated and historical, so no fix
needed; if desired, note the current count is higher.",
653         "severity": "low"
654     },
655     {
656         "claim": "SUPPORTED_LOCALES entries are at i18n/index.ts:18.",
657         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
658         "verdict": "stale",
659         "evidence": "web/src/i18n/index.ts: SUPPORTED_LOCALES is declared at line 25
(= ['en', 'kn', 'hi', 'es', 'da', 'id']), not line 18 (line 18 is the hiCommon import). Line-number
drift only.",
660         "suggestedFix": "Change the citation from i18n/index.ts:18 to
i18n/index.ts:25.",
661         "severity": "low"
662     },
663     {
664         "claim": "The <html lang> sync is at i18n/index.ts:59-65.",
665         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
666         "verdict": "stale",
667         "evidence": "web/src/i18n/index.ts:59-65 is the detection-order block
(order/lookupLocalStorage/etc.); the syncDocumentLang() function for <html lang> begins at line
70.",
668         "suggestedFix": "Repoint the <html lang> citation to ~i18n/index.ts:70
(syncDocumentLang).",
669         "severity": "low"
670     },
671     {
672         "claim": "main.tsx:7-8 is the @fontsource unicode-range-scoped font-load
pattern reused for new fonts.",
673         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49,50",
674         "verdict": "correct",
675         "evidence": "web/src/main.tsx:7-8 = import '@fontsource-variable/noto-sans-
kannada/index.css'; import '@fontsource-variable/noto-sans-devanagari/index.css'; comment at
lines 5-6 describes the unicode-range-scoped self-host.",
676         "suggestedFix": "",
677         "severity": "low"
678     },
679     {
680         "claim": "es needs a 3-form {one, many, other} ICU plural and the tooling
enforces placeholder parity.",
681         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:12,43,114",
682         "verdict": "correct",
683         "evidence": "locales/es/common.json:51 '{count, plural, one {# jugada} many
{# jugadas} other {# jugadas}}' (and lines 59,466). Placeholder-parity guard exists:
web/src/i18n/hygiene.test.ts:53 'describe(\"placeholder parity ? every locale preserves en's
{placeholders}\").'",
684         "suggestedFix": "",
685         "severity": "low"
686     },

```

```

687         {
688             "claim": "The doc is a PEER-OF FRIENDLY_AND_MULTILINGUAL.md, which lives in
docs/archives/.",
689             "docLocation": "docs/MULTILINGUAL_EXPANSION.md:6,17,108",
690             "verdict": "correct",
691             "evidence": "git ls-files shows docs/archives/FRIENDLY_AND_MULTILINGUAL.md
exists (archived via #93 e2570e4). The relative link 'archives/FRIENDLY_AND_MULTILINGUAL.md'
from docs/ resolves correctly.",
692             "suggestedFix": "",
693             "severity": "low"
694         },
695         {
696             "claim": "Doc is indexed in docs/README.md.",
697             "docLocation": "docs/MULTILINGUAL_EXPANSION.md:5",
698             "verdict": "correct",
699             "evidence": "docs/README.md:16 '- **`MULTILINGUAL_EXPANSION.md`** ? proposal
to add **7 more locales** (th/id/da/es/zh-Hans/ja/ko)'.",
700             "suggestedFix": "",
701             "severity": "low"
702         }
703     ]
704 },
705 {
706     "doc": "docs/PER_RALLY_SELECTION.md",
707     "overallStatus": "significant-staleness",
708     "summary": "The doc's MVP thesis and SPA-side deliverables (P0-P5) are accurate
and genuinely shipped, and the future P7/P9/P10 engine endpoints are correctly marked not-built.
But the engine moved on: two whole engine-side items the doc lists as not-done are now shipped
(P6's /api/health AND compile's download_url; P8's /api/videos + db.list_videos), the central
`/api/compile` returns a filepath NOT a URL" claim is now false, the P3/P4/P5 PR numbers were
never backfilled from `(this PR)`, and essentially every [verified backend/main.py:NNN] /
database.py line anchor has drifted off its target. Not an archive candidate ? the doc itself is
still live for the owner-gated M1-M5 work ? but its $4.3 table, $7 tracker, and Foundation
anchors need a refresh pass.",
709     "findings": [
710         {

```

7. assistant (2026-06-22T09:46:38.407Z)

Now I have all the findings for my doc. Let me extract the staleFindings array section. Let me search for the staleFindings array which contains the actual worklist.

8. user (2026-06-22T09:46:41.157Z)

```
1323:     "staleFindings": [
```

9. assistant (2026-06-22T09:46:44.232Z)

The `staleFindings` array is the derived worklist. But the `perDoc[].findings` for my doc already contains the complete set with verdicts. Let me read the staleFindings section to confirm it matches what I extracted, filtering to my doc.

10. user (2026-06-22T09:46:45.933Z)

```

1323     "staleFindings": [
1324         {
1325             "doc": "NEXT_STEPS.md",
1326             "claim": "PRs through #93 merged (master).",
1327             "docLocation": "NEXT_STEPS.md:13",
1328             "verdict": "stale",
1329             "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
1330             "suggestedFix": "Update to 'PRs through #94 merged (master).'",

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1331         "severity": "medium"
1332     },
1333     {
1334         "doc": "README.md",
1335         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
1336         "docLocation": "README.md:9",
1337         "verdict": "stale",
1338         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx; `git
log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c first-run
wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
1339         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the web/
SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
1340         "severity": "high"
1341     },
1342     {
1343         "doc": "README.md",
1344         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
1345         "docLocation": "README.md:4",
1346         "verdict": "stale",
1347         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
1348         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
1349         "severity": "medium"
1350     },
1351     {
1352         "doc": "README.md",
1353         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
1354         "docLocation": "README.md:17",
1355         "verdict": "stale",
1356         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No such
file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs are
deploy/, docs/, locales/, site/, web/ only.",
1357         "suggestedFix": "Mark it as a future/planned directory that does not yet exist
(e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader doesn't
expect the path to be present.",
1358         "severity": "medium"
1359     },
1360     {
1361         "doc": "deploy/DEPLOY_ALPHA.md",
1362         "claim": "header stale deploy live",
1363         "docLocation": "deploy/DEPLOY_ALPHA.md:3",
1364         "verdict": "stale",
1365         "evidence": "droplet/README.md:100 live-verified 2026-06-22",
1366         "suggestedFix": "set header to LIVE",
1367         "severity": "medium"
1368     },
1369     {
1370         "doc": "deploy/DEPLOY_ALPHA.md",
1371         "claim": "host list too narrow",
1372         "docLocation": "deploy/DEPLOY_ALPHA.md:66",
1373         "verdict": "stale",
1374         "evidence": "engine.env.example adds loopback",
1375         "suggestedFix": "add localhost",
1376         "severity": "low"
1377     },
1378     {

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1379         "doc": "docs/ANNOTATE_AND_TRAIN.md",
1380         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last updated:
2026-06-20.",
1381         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
1382         "verdict": "stale",
1383         "evidence": "git log --follow shows the marketing-page work merged: P0 doc (#15,
6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per §7, P0-P4 are all
marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN PROGRESS->DONE;
with P0-P4 shipped this is at least IN PROGRESS.",
1384         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain) and
bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
1385         "severity": "medium"
1386     },
1387     {
1388         "doc": "docs/ANNOTATE_AND_TRAIN.md",
1389         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
1390         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
1391         "verdict": "stale",
1392         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots' committed
site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133 bytes); own-
model.html:111-112 references them as the real VLC-UI shots (no people/footage). The annotator-
UI half of P5 is done and merged, not Todo.",
1393         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
1394         "severity": "high"
1395     },
1396     {
1397         "doc": "docs/ANNOTATE_AND_TRAIN.md",
1398         "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
1399         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
1400         "verdict": "stale",
1401         "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
1402         "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to docs/README.md's
'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
1403         "severity": "high"
1404     },
1405     {
1406         "doc": "docs/ANNOTATE_AND_TRAIN.md",
1407         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
1408         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
1409         "verdict": "stale",
1410         "evidence": "grep on web/package.json: no react-router dep (claim 'no router'
still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and conditionally
renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88). The 'one
static screen' framing predates the wizard.",
1411         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run SetupWizard +
the ingest panel' so the SPA surface description matches the post-#88 code.",
1412         "severity": "low"
1413     },
1414     {
1415         "doc": "docs/ANNOTATE_AND_TRAIN.md",
1416         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",
1417         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
1418         "verdict": "stale",
1419         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md already
exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13' /
'cross-link once PR #13 merges' phrasing is outdated.",

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1420         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
1421         "severity": "low"
1422     },
1423     {
1424         "doc": "docs/DRY_RUN_CUJ.md",
1425         "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
1426         "docLocation": "docs/DRY_RUN_CUJ.md:3",
1427         "verdict": "stale",
1428         "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD $6
boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57). A
DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
1429         "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21) and
drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
1430         "severity": "medium"
1431     },
1432     {
1433         "doc": "docs/DRY_RUN_CUJ.md",
1434         "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
1435         "docLocation": "docs/DRY_RUN_CUJ.md:11",
1436         "verdict": "stale",
1437         "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =
[\"en\", \"kn\", \"hi\", \"es\", \"da\", \"id\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
1438         "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)' where it
describes the current switcher; the Hindi/Kannada run results in $7 stay accurate as historical
record.",
1439         "severity": "low"
1440     },
1441     {
1442         "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
1443         "claim": "$3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; $4.6 Reuse map lists it under 'Net-new'; $9 T7 (capabilities
+ health + banner) is ? Todo.",
1444         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
1445         "verdict": "stale",
1446         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
1447         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out of
$4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (??), leaving only
the SPA banner pending.",
1448         "severity": "high"
1449     },
1450     {
1451         "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
1452         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS `*`
`[verified backend/main.py:42]` ? structural safety only because the engine binds 127.0.0.1.",
1453         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
1454         "verdict": "stale",
1455         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email`/`owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
1456         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable
(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
1457         "severity": "high"
1458     },

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1459     {
1460         "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
1461         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
1462         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
1463         "verdict": "stale",
1464         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
1465         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
1466         "severity": "medium"
1467     },
1468     {
1469         "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
1470         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
1471         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
1472         "verdict": "stale",
1473         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the droplet
(2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-fast,
Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live; full
DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
1474         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
1475         "severity": "high"
1476     },
1477     {
1478         "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
1479         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with $9
the source of truth.",
1480         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
1481         "verdict": "stale",
1482         "evidence": "khelsutra git log shows merges through #94 (master HEAD 50b2bd7)
and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'. Multiple $9
rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on master).",
1483         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN PROGRESS;
refresh $9 rows to current reality.",
1484         "severity": "medium"
1485     },
1486     {
1487         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1488         "claim": "EX3 ('site/ marketing parity for the 7') is ? and 'Blocked on B3'; and
$1/$4/$6 say site localization 'depends on FRIENDLY B3 landing first ? until then this is SPA-
only.'",
1489         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
1490         "verdict": "stale",
1491         "evidence": "FRIENDLY B3 shipped: git log 1b1fa7b 'feat(web+site): B3 ?
trilingual marketing site via shared t()/data-i18n shim (#70)'. Site es/da/id parity also
shipped: cf59fbf '#87'. git ls-files shows site/public/locales/{es,da,id}/common.json present.
The dependency is satisfied and the Phase-A trio is no longer SPA-only.",
1492         "suggestedFix": "Update EX3 to reflect that FRIENDLY B3 landed (#70) and the
Phase-A es/da/id site parity shipped (#87) ? mark the es/da/id portion ? (site has those
catalogs) with only th/zh/ja/ko site parity remaining; drop the 'until then this is SPA-only'
caveat for es/da/id in $1/$4/$6.",
1493         "severity": "medium"
1494     },
1495     {
1496         "doc": "docs/MULTILINGUAL_EXPANSION.md",

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1497         "claim": "$10 References cites internal 'docs/UI_PROPOSAL.md'.",
1498         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
1499         "verdict": "stale",
1500         "evidence": "git ls-files | grep -i UI_PROPOSAL returns nothing (exit 1) ? no
such file exists anywhere in the repo.",
1501         "suggestedFix": "Remove the docs/UI_PROPOSAL.md reference or repoint it to the
correct doc (the engine repo has UI docs; this repo does not contain UI_PROPOSAL.md).",
1502         "severity": "low"
1503     },
1504     {
1505         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1506         "claim": "SUPPORTED_LOCALES entries are at i18n/index.ts:18.",
1507         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
1508         "verdict": "stale",
1509         "evidence": "web/src/i18n/index.ts: SUPPORTED_LOCALES is declared at line 25 (=
['en','kn','hi','es','da','id']), not line 18 (line 18 is the hiCommon import). Line-number
drift only.",
1510         "suggestedFix": "Change the citation from i18n/index.ts:18 to
i18n/index.ts:25.",
1511         "severity": "low"
1512     },
1513     {
1514         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1515         "claim": "The <html lang> sync is at i18n/index.ts:59-65.",
1516         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
1517         "verdict": "stale",
1518         "evidence": "web/src/i18n/index.ts:59-65 is the detection-order block
(order/lookupLocalStorage/etc.); the syncDocumentLang() function for <html lang> begins at line
70.",
1519         "suggestedFix": "Repoint the <html lang> citation to ~i18n/index.ts:70
(syncDocumentLang).",
1520         "severity": "low"
1521     },
1522     {
1523         "doc": "docs/PER_RALLY_SELECTION.md",
1524         "claim": "P6 (engine repo) ? GET /api/health + offline banner; /api/compile
returns download_url ? Status ? (Todo, not built); $4.3 table marks GET /api/health as 'new' and
the download_url addition as 'modify' (not yet done).",
1525         "docLocation": "docs/PER_RALLY_SELECTION.md:133 (P7 tracker row P6), :83-84
($4.3 table)",
1526         "verdict": "stale",
1527         "evidence": "backend/main.py:307 `@app.get('/api/health')` is fully
implemented; backend/main.py compile (~line 769) returns
`{"status\":\"success\", \"filepath\": \"out_path\", \"download_url\": \"download_url\"}` where
download_url = `/api/highlights/{video_id}/{out_name}`. Both P6 items are shipped in the
engine.",
1528         "suggestedFix": "Flip P6 to ? (or split: health=?, download_url=?) and update
$4.3 status from 'new'/'modify' to 'exists'. Note in the doc that the engine now returns
download_url so the SPA no longer needs to hand-build the URL (though it still can).",
1529         "severity": "high"
1530     },
1531     {
1532         "doc": "docs/PER_RALLY_SELECTION.md",
1533         "claim": "Compile returns a server-local filepath, NOT a URL, so the SPA
constructs this download URL itself (repeated as a core caveat in $0 TL;DR, $3 Foundation, $4.2
data flow, D11/risk table, changelog).",
1534         "docLocation": "docs/PER_RALLY_SELECTION.md:12, :44, :73, :116",
1535         "verdict": "stale",
1536         "evidence": "backend/main.py compile endpoint now returns `download_url`
alongside `filepath`: `download_url =
f\"/api/highlights/{quote(req.video_id)}/{quote(out_name)}\"; return
{\"status\":\"success\", \"filepath\": \"out_path\", \"download_url\": \"download_url\"}`. The 'filepath not
a URL' premise that the whole 'SPA must build the URL' design rests on is no longer the full
picture.",
1537         "suggestedFix": "Update the caveat to: compile returns BOTH a server-local

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filepath and a ready-to-use download_url (/api/highlights/...). The SPA may use download_url
directly; the hand-built-URL fallback is now optional, not required.",
1538         "severity": "high"
1539     },
1540     {
1541         "doc": "docs/PER_RALLY_SELECTION.md",
1542         "claim": "P8 ? GET /api/videos + identity scoping ? Status ? owner, ? (not
built); §4.3 marks 'GET /api/videos' as 'new', noting 'only get_video(id) exists,
database.py:418'." ,
1543         "docLocation": "docs/PER_RALLY_SELECTION.md:135 (P8 row), :86 (§4.3)",
1544         "verdict": "stale",
1545         "evidence": "backend/main.py:374 `@app.get(\"/api/videos\")` ?
`db_instance.list_videos(limit=limit)`; backend/infrastructure/database.py:545 `def
list_videos(...)` . The endpoint and the DB enumerate both exist (added per PACKAGING_PLAN.md
P1). The doc's 'only get_video(id) exists' claim and the 'new' status are wrong; identity
scoping may still be pending but the listing endpoint shipped.",
1546         "suggestedFix": "Mark the GET /api/videos endpoint as 'exists' in §4.3; in P8
note that the videos-list endpoint already shipped and only the identity/owner-scoping layer
remains gated.",
1547         "severity": "high"
1548     },
1549     {
1550         "doc": "docs/PER_RALLY_SELECTION.md",
1551         "claim": "P3/P4/P5 tracker rows show PR = '(this PR)' as the merge reference.",
1552         "docLocation": "docs/PER_RALLY_SELECTION.md:130-132",
1553         "verdict": "stale",
1554         "evidence": "git log: P3 merged as #28 (d537d19), P4 as #30 (aff8de8), P5 as #31
(8bf200d). The '(this PR)' placeholders were never backfilled, so a reader cannot find the
actual PRs.",
1555         "suggestedFix": "Replace '(this PR)' with [#28], [#30], [#31] for P3/P4/P5
respectively.",
1556         "severity": "medium"
1557     },
1558     {
1559         "doc": "docs/PER_RALLY_SELECTION.md",
1560         "claim": "Engine line anchors in [verified ...] tags: config at
backend/main.py:121-123; jobs :274-293; highlights download :307-328; query :331-340; compile +
floor :342-396; play_segments database.py:124-135; dead correction cols :166-168; no list-videos
:418; query read :427-475.",
1561         "docLocation": "docs/PER_RALLY_SELECTION.md:38-46, :163 (§10 anchors)",
1562         "verdict": "stale",
1563         "evidence": "Live engine: /api/config is at main.py:297 (not 121-123);
/api/jobs/{job_id} at :637 (not 274-293); /api/highlights at :670 (not 307-328); /api/query at
:694 (not 331-340); /api/compile at :705 (not 342-396); play_segments CREATE TABLE at
database.py:186 (not 124-135); get_video at :536 (not 418). The endpoints/contracts still EXIST
and behave as described, but every cited line number has drifted because the engine ships PRs
independently.",
1564         "suggestedFix": "Re-run the anchors against current engine HEAD and update line
numbers, or replace brittle line numbers with symbol names (e.g. main.py::get_config,
database.py::list_videos). The sibling DRY_RUN_CUJ.md §8 already warns that engine-surface
claims drift between sessions.",
1565         "severity": "medium"
1566     },
1567     {
1568         "doc": "docs/PER_RALLY_SELECTION.md",
1569         "claim": "References §0.6 / §10 cite 'REQUEST_ACCESS_FORM.md' (repo-root style)
as a peer doc.",
1570         "docLocation": "docs/PER_RALLY_SELECTION.md:6, :164",
1571         "verdict": "stale",
1572         "evidence": "git ls-files shows the file is at docs/REQUEST_ACCESS_FORM.md, not
at repo root. The bare 'REQUEST_ACCESS_FORM.md' reference is path-imprecise (it now lives under
docs/).",
1573         "suggestedFix": "Reference it as docs/REQUEST_ACCESS_FORM.md.",
1574         "severity": "low"
1575     },

```

```

1576     {
1577         "doc": "docs/PORTABILITY.md",
1578         "claim": "`node --test src/` runs BOTH paths.",
1579         "docLocation": "docs/PORTABILITY.md:25",
1580         "verdict": "stale",
1581         "evidence": "package.json `test` is: `node --test` (no src/ arg) is what CI
runs (.github/workflows/test.yml:25 `npm test`). The split targets are test:cf / test:vm. `node
--test src/` is not the command CI uses; the bare `node --test` discovers and runs all
*.test.js.",
1582         "suggestedFix": "Change to: `npm test` (node --test) runs BOTH paths, and CI
runs it on every PR.",
1583         "severity": "low"
1584     },
1585     {
1586         "doc": "docs/PORTABILITY.md",
1587         "claim": "VM deploy runbook: Node 22+ only ? no npm install (node:sqlite is
built in).",
1588         "docLocation": "docs/PORTABILITY.md:64",
1589         "verdict": "stale",
1590         "evidence": "Internally inconsistent with the doc's own provisioning section
(line 40: Node >=24 'so node:sqlite is unflagged') and the repo: provision-vm.sh NODE_MAJOR
default 24 + 'must be >= 24', .github/workflows/test.yml:17 node-version 24, server.js:1 also
says 'Node 22+'. node:sqlite was flagged/experimental before Node ~22.5 and only stabilized
later; a reader running the runbook on Node 22 may hit a flagged/unavailable node:sqlite. The
authoritative requirement in code is >=24.",
1591         "suggestedFix": "Change 'Node 22+ only' to 'Node 24+ only (node:sqlite is
unflagged at >=24)' in both the runbook (line 64) and server.js:1, to match provision-vm.sh /
CI.",
1592         "severity": "medium"
1593     },
1594     {
1595         "doc": "docs/README.md",
1596         "claim": "MULTILINGUAL_EXPANSION.md is a 'proposal to add 7 more locales
(th/id/da/es/zh-Hans/ja/ko) ... DRAFT ? owner-gated (approval to start), [#58].'",
1597         "docLocation": "docs/README.md:16-18",
1598         "verdict": "stale",
1599         "evidence": "docs/MULTILINGUAL_EXPANSION.md:3 status header reads 'IN PROGRESS ?
Phase A (es/da/id) shipped (#67); Phases B (Thai) / C (CJK) owner-gated'. Git confirms PR #67
'feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs'. site/public/locales/
now contains es, da, id directories and web/src/i18n/index.ts:25 lists SUPPORTED_LOCALES =
en, kn, hi, es, da, id. So es/da/id are no longer 'to add', and the doc is not a not-yet-started
DRAFT.",
1600         "suggestedFix": "Update to: '**`MULTILINGUAL_EXPANSION.md`' ? `IN PROGRESS`:
Phase A (es/da/id) shipped ([#67]); Thai (B) / CJK (C) remain owner-gated. Adds locales beyond
the en/kn/hi foundation; peer of the archived `FRIENDLY_AND_MULTILINGUAL.md`.'",
1601         "severity": "high"
1602     },
1603     {
1604         "doc": "docs/README.md",
1605         "claim": "The index lists every tracked doc in this repo (DRY_RUN_CUJ,
FRIENDLY_AND_MULTILINGUAL, MULTILINGUAL_EXPANSION, LOCAL_APPLIANCE_ARCHITECTURE,
PER_RALLY_SELECTION, REQUEST_ACCESS_FORM, PORTABILITY).",
1606         "docLocation": "docs/README.md:3-27",
1607         "verdict": "stale",
1608         "evidence": "git ls-files docs/ also returns docs/ANNOTATE_AND_TRAIN.md, which
is NOT referenced anywhere in README.md. That doc's own header (docs/ANNOTATE_AND_TRAIN.md:5)
states its tracking anchor is 'index in docs/README.md', so it expects to be listed here.",
1609         "suggestedFix": "Add a 'Tracked work' entry, e.g.: '**`ANNOTATE_AND_TRAIN.md`'
? annotate-your-footage ? train-your-own-model USP pages (rally-annotator + engine `worker
train`). `DRAFT` ? owner-gated on scope.",
1610         "severity": "medium"
1611     },
1612     {
1613         "doc": "docs/REQUEST_ACCESS_FORM.md",
1614         "claim": "Frontend lives in `site/index.html` (the header/hero buttons scroll to

```

```

#join; #join becomes the form).",
1615         "docLocation": "docs/REQUEST_ACCESS_FORM.md:58",
1616         "verdict": "stale",
1617         "evidence": "`ls site/index.html` -> exit 2 (no such file). The frontend is git-
tracked as site/public/index.html (git ls-files), which contains the #join section
(public/index.html:524) and the href=`\"#join\"` CTAs (lines 269,282,284). The file was moved by
PR #3 (git show f699847 --stat: `site/{ => public}/index.html`) on 2026-06-20 ? the same day as
the doc's 'Last updated', so the path reference was never updated.",
1618         "suggestedFix": "Change `site/index.html` to `site/public/index.html` in §4 (the
#join + form behavior it describes is otherwise correct).",
1619         "severity": "medium"
1620     },
1621     {
1622         "doc": "docs/REQUEST_ACCESS_FORM.md",
1623         "claim": "The hidden honeypot field is named `hp` (anti-spam; if filled ->
silently drop).",
1624         "docLocation": "docs/REQUEST_ACCESS_FORM.md:55",
1625         "verdict": "stale",
1626         "evidence": "site/public/index.html:602 `

```

```

and note the test files (index/server/assets/deploy-config/i18n.shim/locales-drift/ux.dom +
layout.browser.mjs).",
1655         "severity": "medium"
1656     },
1657     {
1658         "doc": "site/README.md",
1659         "claim": "The site is just 'a static landing page + /capture checklist plus a
small Worker' ? Cloudflare-Worker-only; server.js / VM host / droplet are not mentioned at
all.",
1660         "docLocation": "site/README.md:1-12",
1661         "verdict": "stale",
1662         "evidence": "site/server.js:1-14 is an 'Always-on VM host for the KhelSutra site
... the portability fallback if Cloudflare acts out ? see docs/PORTABILITY.md', sharing the same
handleRequest from src/index.js. docs/PORTABILITY.md exists. The site test run shows a full '[VM
host]' suite (POST persists to SQLite, serves /own-model, flyers, locales). The README has zero
mention of server.js, the VM host, the droplet, or PORTABILITY.md.",
1663         "suggestedFix": "Add a short section: the same handler runs in two places ? the
Cloudflare Worker (src/index.js, this README's focus) and an always-on VM host (server.js, the
portability fallback; see docs/PORTABILITY.md), used by the alpha droplet.",
1664         "severity": "medium"
1665     },
1666     {
1667         "doc": "site/README.md",
1668         "claim": "public/ ? static assets served as-is (index.html, capture.html,
og.png, robots.txt).",
1669         "docLocation": "site/README.md:7",
1670         "verdict": "stale",
1671         "evidence": "git ls-files site/public/ also lists own-model.html,
flyer.html/flyer-kn.html/flyer-hi.html, i18n.js, img/ (studio/annotate/flyer-qr), video/tool-
cu.mp4, and locales/{en,kn,hi,es,da,id}/common.json ? a 6-language i18n + flyers + SPA-video
surface absent from the README's 4-file list.",
1672         "suggestedFix": "Broaden the public/ description to mention the localized pages
+ locales/ (6 langs), the flyers, own-model.html, img/, and video/, not just the original four
files.",
1673         "severity": "low"
1674     },
1675     {
1676         "doc": "site/README.md",
1677         "claim": "npm test # node --test src/ (no install needed)",
1678         "docLocation": "site/README.md:16",
1679         "verdict": "stale",
1680         "evidence": "site/package.json scripts.test = 'node --test' (no src/ path
argument). Running it executes 86 tests across src/ and other dirs (? tests 86, pass 86).",
1681         "suggestedFix": "Change the annotation to '# node --test (86 tests, no install
needed)'.",
1682         "severity": "low"
1683     },
1684     {
1685         "doc": "site/README.md",
1686         "claim": "src/index.js ? the Worker: serves public/ and handles POST
/api/request-access -> D1.",
1687         "docLocation": "site/README.md:8",
1688         "verdict": "stale",
1689         "evidence": "site/src/index.js:84 returns env.ASSETS.fetch(request) for non-form
paths ? on Cloudflare the static assets are served by the [assets] binding (wrangler.toml:12-14
ASSETS), not by Worker code. The Worker only owns the /api/request-access path; it delegates
'everything else: static assets'.",
1690         "suggestedFix": "Reword to 'handles POST /api/request-access -> D1 and delegates
all other paths to the Cloudflare static-assets binding (ASSETS)'.",
1691         "severity": "low"
1692     },
1693     {
1694         "doc": "web/README.md",
1695         "claim": "`src/App.tsx` ? the console shell (screens land in M2+).",
1696         "docLocation": "web/README.md:30",

```

```

1697         "verdict": "stale",
1698         "evidence": "web/src/App.tsx is a complete screen, not a shell: it renders
RallyGrid, QueryBar, TapFilters, SelectionFooter, LanguageSwitcher, IngestPanel, SetupWizard,
CompileResult (imports App.tsx:3-16) and orchestrates listRallies/compileReel against the engine
(App.tsx:80,150). Last touched by PR #88 'P2c first-run setup wizard' (git log --
web/src/App.tsx), long past M2. The README was last modified at PR #14 (git log --
web/README.md), so this line predates the entire screen build-out.",
1699         "suggestedFix": "Replace with something like: \"`src/App.tsx` ? the operator
console: per-rally selection (RallyGrid/QueryBar/TapFilters/SelectionFooter), in-UI ingest
(IngestPanel), the P2c first-run wizard (SetupWizard) and reel compile, in 6 languages.\"\"",
1700         "severity": "high"
1701     },
1702     {
1703         "doc": "web/README.md",
1704         "claim": "Layout section lists only src/api/client.ts, src/api/types.ts,
src/lib/log.ts, src/App.tsx as the SPA's structure.",
1705         "docLocation": "web/README.md:24-30",
1706         "verdict": "stale",
1707         "evidence": "git ls-files web/ shows the SPA has grown a full tree the Layout
omits: src/components/ (9 non-test components incl. RallyGrid, QueryBar, SetupWizard,
IngestPanel, LanguageSwitcher), src/hooks/ (useIngestJob, useRallySelection, useUpload),
src/i18n/ (6-locale catalog: en/kn/hi/es/da/id per grep of src/i18n/index.ts), and most of
src/lib/ (query, selection, floor, format, errors, errorText). The README froze at the #14
scaffold (git log -- web/README.md) when only these four files existed.",
1708         "suggestedFix": "Add the components/, hooks/, i18n/ (6 languages) and remaining
lib/ entries so the Layout reflects the shipped SPA, not just the P1 scaffold.",
1709         "severity": "medium"
1710     },
1711     {
1712         "doc": "web/README.md",
1713         "claim": "\"src/api/client.ts` ? typed engine client (`listRallies` /
`compileReel` / `highlightUrl` / `getConfig`)",
1714         "docLocation": "web/README.md:25-26",
1715         "verdict": "stale",
1716         "evidence": "All four named exports exist
(web/src/api/client.ts:101,111,128,133), so the claim is not wrong ? but the client now also
exports processVideo (:85), getJob (:96), getCapabilities (:138), minShotCount (:143) and the
chunked-upload family uploadInit/uploadPart/uploadComplete/uploadAbort/uploadFile (:152-245).
The four-function summary undersells the current surface.",
1717         "suggestedFix": "Append `+ processVideo / getJob / getCapabilities / chunked
uploadFile` (or note the list is illustrative) so the client description isn't read as
exhaustive.",
1718         "severity": "low"
1719     }
1720 ]
1721 }
1722 }

```

11. assistant (2026-06-22T09:46:52.569Z)

I now have my complete worklist. There are 5 stale findings for
`docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md`:

1. `/api/capabilities` now exists on engine master (\$3, \$4.6, \$9 T7) ? HIGH
2. "zero app-layer auth and CORS ``" now stale ? engine ships Host-guard + edge-auth (\$0, \$6 risk, D1) ? HIGH
3. Native WASB now on master, not branch-only (\$3, \$4.3) ? MEDIUM
4. Alpha deploy is LIVE 2026-06-22 (\$9, \$11 DoD) ? HIGH
5. Status header date/lifecycle stale (line 3) ? MEDIUM

Let me re-verify each against the live engine repo (origin/master). Let me run the verification commands in parallel.

12. user (2026-06-22T09:46:55.268Z)

```

=== capabilities ===
origin/master:backend/main.py:340:@app.get("/api/capabilities")
=== health ===
origin/master:backend/main.py:307:@app.get("/api/health")
=== version ===
origin/master:backend/main.py:334:@app.get("/api/version")

```

13. user (2026-06-22T09:46:57.694Z)

```

=== cors ===
origin/master:backend/main.py:66:def _cors_origins_from_env(env: Optional[Dict[str, str]] =
None) -> List[str]:
origin/master:backend/main.py:76:# CORS for the web UI. allow_origins='' with
allow_credentials=True is rejected by browsers per the
origin/master:backend/main.py:82:     allow_origins=_cors_origins_from_env(),
origin/master:backend/main.py:83:     allow_credentials=False,
=== trustedhost ===
origin/master:backend/main.py:33:from starlette.middleware.trustedhost import
TrustedHostMiddleware
origin/master:backend/main.py:100:     # IPv4 loopback (the default bind) + the TestClient host.
NB: Starlette's TrustedHostMiddleware
origin/master:backend/main.py:110:     TrustedHostMiddleware,
=== auth file ===
100644 blob b824d9d0d9e82cfdfbade589697323088feb77f6      backend/api/auth.py
100644 blob f8c4e114435fd6152dadb5fa573063b9ff42e172      backend/config/startup.py
=== cf access ===

```

14. assistant (2026-06-22T09:47:00.649Z)

The Cf-Access grep on backend/ returned no hits. Let me verify the exact location of the Cf-Access header trust and the native WASB / startup posture.

15. user (2026-06-22T09:47:02.441Z)

```

=== cf access broad ===
origin/master:backend/api/auth.py:1:"""Cloudflare Access edge-auth ? verify the Cf-Access JWT,
derive the tenant ``user_id`` (M9, D9).
origin/master:backend/api/auth.py:5:``Cf-Access-Jwt-Assertion`` header is **required** and
**verified** (RS256) against the team JWKS ?
origin/master:backend/api/auth.py:23:CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
origin/master:backend/api/auth.py:31:     """A required Cf-Access token was missing or failed
verification ? the caller returns 401/403."""
origin/master:backend/api/auth.py:67:def verify_cf_access_token(token: str, *, team_domain: str,
aud: str, jwks: dict) -> str:
origin/master:backend/api/auth.py:68:     """Verify a Cf-Access JWT against ``jwks`` (RS256;
checks sig + exp + issuer + audience) and
origin/master:backend/api/auth.py:94:         raise EdgeAuthError(f"Cf-Access token verification
failed: {exc}") from exc
origin/master:backend/api/auth.py:107:     Enabled ? require + verify the Cf-Access header; raise
``EdgeAuthError`` (? 401/403) on any miss.
origin/master:backend/api/auth.py:117:     aud = (getattr(sec, "cf_access_aud", "") or
"").strip()
origin/master:backend/api/auth.py:120:         "edge_auth_enabled but security.cf_team_domain
/ cf_access_aud are not configured")
=== owner_id ===

```

16. user (2026-06-22T09:47:05.056Z)

```

=== native_wasb on master ===
100644 blob 43037b6803786a7cd7e40d7c21bf678585949e86
backend/pipeline/detectors/native_wasb_runner.py
100644 blob 32df561632405388eb22ef8537b09b6f6dc496b0      tests/test_native_wasb_runner.py
=== trajectory still raises ===
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:56:         raise

```

```

NotImplementedError
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:58:     def
_build_native_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str, Any]]:
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:61:         raise
NotImplementedError(
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:73:
``_build_native_runner``. Anything else (``"auto"``) delegates to ``_build_runner``
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:86:         return
self._build_native_runner(idx_cfg)
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:133:         ``FusionSegmenter``
overrides this to apply Gate A/B when thresholds are raised.
origin/master:backend/pipeline/segmenters/trajectory_hybrid.py:196:     except
NotImplementedError as e:
=== fusion builds it ===
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:76:     def _build_native_runner(self,
idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str, Any]]:
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:84:         from
backend.pipeline.detectors.native_wasb_runner import NativeWasbRunner, NativeWasbConfig
origin/master:backend/pipeline/segmenters/fusion_hybrid.py:86:         return
NativeWasbRunner(cfg), {"weights_path": cfg.weights_path, "detector_impl": "native",

```

17. assistant (2026-06-22T09:47:11.696Z)

All engine-side claims confirmed against origin/master:

- `/api/capabilities`` at `main.py:340`, `/api/health`` at `:307` ? both exist.
- CORS: ``allow_origins=cors_origins_from_env()``` (default ``*```, env-lockable), ``allow_credentials=False``; `TrustedHostMiddleware` exists.
- Edge auth: ``backend/api/auth.py`` exists, verifies `Cf-Access-JWT`` (``Cf-Access-Jwt-Assertion``), ``enforce_startup_checks`` in `startup.py`.
- `native_wasb_runner.py`` + test on master; `trajectory_hybrid`_build_native_runner`` still raises `NotImplementedError`` (line 61); `fusion_hybrid`` builds `NativeWasbRunner`` (line 86).

Note: the audit said the engine trusts ``Cf-Access-Authenticated-User-Email``, but the actual header is ``Cf-Access-Jwt-Assertion`` (RS256-verified JWT), and ``owner_id`` is not in `auth.py`. I'll word the doc fix accurately to match what's actually on master rather than copying the audit's slightly-off phrasing. Now let me verify the khelsutra-side deploy-LIVE claim in ``deploy/droplet/README.md``.

18. user (2026-06-22T09:47:12.686Z)

```

96      **The shareable CUJ:** open `app.khelsutra.guru` ? sign in (email OTP) ? **upload a
video** (lands in the
97      jail) ? Process ? pick rallies ? Build ? download the reel. *(Pasting a `gs://` URI is
rejected by the
98      jail on the exposed box ? upload is the alpha path; see `../DEPLOY_ALPHA.md` $7.)*
99
100     > **? Live-verified on the droplet (2026-06-22).** Engine installed + booted on
`168.144.126.176`; full
101     > suite **1313 passed** + ruff/mypy clean on the box; boot-posture fail-fast, the Host-
guard `400?200`,
102     > the path-jail (`gs://`/etc/passwd`/`. ` ? 400), the T4 upload?jail, and the edge-
auth `401`-gate all
103     > exercised live. Tunnel + Access + Pages are up; the full **DNS ? Cloudflare ? Access ?
tunnel ? engine**
104     > chain is confirmed (`/api/health` = 200 through an Access session; unauthenticated ?
302 to the login).
105     > Two things still need a human: the rotated **Gemini key** (`/etc/khelsutra/engine.env`
? restart) and a
106     > **browser one-time-PIN login** to exercise the upload?reel CUJ.
107     >
108     > **? Two gotchas the live run surfaced:**
109     > - **`indexing.default_segmenter` MUST be `gemini` on a CPU box** (set in
`config.json.example`). The SPA
110     > sends no segmenter, so the engine uses its default ? the shipped default
`yolo_hybrid` needs torch/GPU

```

```
111     > and every job fails `No module named 'torch'` without this.
112     > - **Without a valid `GEMINI_API_KEY`, a job finishes `done` with 0 rallies** ? the
missing-key error lands
113     > only in the engine log, not the job status. Set the key before sharing.
114
```

19. assistant (2026-06-22T09:47:18.425Z)

The droplet-LIVE claim is confirmed in the khelsutra repo. Now let me look at the exact doc lines I need to edit. Let me re-read the precise target lines: §3 line 45 (native wasb) and 49 (absent), the TL;DR line 14, D1 line 30, §6 risk line 128, §4.3 line 96 (native), §4.6 line 113, §9 T7 line 164, §9 alpha-deploy line 175, §11 DoD lines 182-183, and the header line 3. I have these from the full read already. Let me apply the surgical edits.

****Finding 5 ? Status header (line 3):**** bump "Last updated" and consider DRAFT?IN PROGRESS. The doc has shipped rows (T4, T11, T12 ?) and the alpha is live, so IN PROGRESS is honest.

20. user (2026-06-22T09:47:20.952Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

21. assistant (2026-06-22T09:47:23.360Z)

****Finding 2 ? TL;DR line 14 (the "single most important caveat"):**** the engine is no longer "zero app-layer auth and CORS `\*`". Reword to match what's actually on master.

22. user (2026-06-22T09:47:30.052Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

23. assistant (2026-06-22T09:47:32.997Z)

****Finding 3 ? §3 line 45 (native WASB in-flight on branch):**** now on master.

24. user (2026-06-22T09:47:37.016Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

25. assistant (2026-06-22T09:47:40.039Z)

****Finding 1 ? §3 line 49 (Absent net-new list):**** `/api/capabilities` now exists on master; remove it from the "Absent" list.

26. user (2026-06-22T09:47:43.882Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

27. assistant (2026-06-22T09:47:47.497Z)

****Finding 3 ? §4.3 line 96 (compute matrix, native WASB "in-flight"):**** update the "(in-flight)" framing.

28. user (2026-06-22T09:47:52.491Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

29. assistant (2026-06-22T09:47:58.951Z)

****Finding 2 ? D1 line 30 and §6 risk line 128:**** these both anchor to `main.py:42` for "CORS `\*`". The audit's docLocation lists lines 14,30,103,128 for this finding. Line 30 (D1) says "CORS `\*` is moot" ? that's still substantively true (the proxy makes it moot), but the `[verified main.py:638]` anchor there is for the bind, which is fine. Let me check line 103 (§4.4) and line 128 (§6 risk). Line 103 doesn't reference main.py:42. The key stale anchor is line 128 `[main.py:42]`. Let me fix that risk-row anchor.

30. user (2026-06-22T09:48:02.147Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

31. assistant (2026-06-22T09:48:06.909Z)

****Finding 1 ? §4.6 Reuse map line 113:**** move `/api/capabilities` out of "Net-new". Note `GET /api/health` is already correctly in the reused-list context (and was noted as existing at line 47). The Net-new list has both `GET /api/capabilities` and `GET /api/health`. Health already exists (§3 line 47 says so), and now capabilities does too. Let me move both out, keeping `/api/videos` and `/api/reels` as still-net-new.

32. user (2026-06-22T09:48:10.369Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

33. assistant (2026-06-22T09:48:14.275Z)

****Finding 1 ? §9 T7 line 164:**** mark the engine half (capabilities+health) as shipped, leaving the SPA banner pending. Change ? ? ?.

34. user (2026-06-22T09:48:16.703Z)

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35. assistant (2026-06-22T09:48:21.418Z)

Also fix the §6 risk-table row "New engine endpoints" (line 133) which says capabilities/health "don't exist" ? now partially stale. Let me update it to reflect capabilities/health shipped.

36. user (2026-06-22T09:48:23.740Z)

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37. assistant (2026-06-22T09:48:26.942Z)

****Finding 4 ? §9 alpha-deploy block line 175:**** the alpha stand-up is now LIVE (2026-06-22). Update the "actual stand-up ... is the owner step" framing.

38. user (2026-06-22T09:48:33.550Z)

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39. assistant (2026-06-22T09:48:37.048Z)

****Finding 4 ? §11 DoD boxes 1 and 2 (lines 182-183):**** these are now tested live. Update them to reflect the live verification while keeping the residual owner steps honest (DoD box 1 CUJ-1 needs the browser-OTP login to fully exercise upload?reel, so I'll qualify rather than fully tick).

40. user (2026-06-22T09:48:41.123Z)

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41. assistant (2026-06-22T09:48:47.610Z)

Now let me also check the §3 line 47 health-endpoint anchor and the §12 References section, which carry the stale `main.py:42` CORS anchor and other drifted line numbers. The audit findings didn't list §12 explicitly, so I'll stay within scope ? but line 47's `[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]` for /api/health: on origin/master /api/health is at :307, not :127. That anchor drift wasn't called out by the audit. To avoid scope creep beyond the findings, I'll leave §12 and the §3 health line-number alone (the audit flagged capabilities/auth/native/deploy/header, not the health line-number drift).

Let me verify the §1 line and §3 line 47 don't independently re-assert "zero auth" in a way that contradicts my fix. Let me re-read the relevant edited regions to confirm coherence.

42. user (2026-06-22T09:48:48.265Z)

```
102
103     ### 4.4 Auth / exposure (non-negotiable invariant)
104     No engine auth, by design; the gate is external. **Invariant (all three must hold):**
engine bound to `127.0.0.1` + firewall closes `8000` + edge gate up. Go-live checklist
`[HOSTING_PLAN.md:102-108]`: Access/basicauth ON; **rotate the chat-shared `GEMINI_API_KEY`**;
```

```
set `security.allowed_video_root`; rate-limit; per-user quota.
105
```

43. assistant (2026-06-22T09:48:56.040Z)

§4.4 line 104 says "No engine auth, by design" ? this is now partially stale since the engine ships an optional edge-auth integration. However, the audit's finding #2 docLocations were lines 14,30,103,128 ? line 103 is the §4.4 heading. The substance of §4.4 (the gate is the boundary; the invariant of bind+firewall+gate) remains the design intent and is still correct as the operative posture. The audit's suggestedFix focused on the "zero app-layer auth" literal claim in TL;DR and the risk row, which I've fixed. Line 104's "No engine auth, by design" is a design-stance statement, not a verified-fact claim, and the audit did not flag line 104 specifically. To stay surgical and avoid contradicting the doc's design framing, I'll add a brief honest note rather than rewrite the stance, since the engine now *does* ship an opt-in edge-auth layer the doc's own TL;DR now references.

44. user (2026-06-22T09:48:58.820Z)

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45. assistant (2026-06-22T09:49:06.854Z)

All five stale findings are addressed. Let me do a final coherence check on the changelog section to optionally record this refresh ? but the audit didn't request a changelog entry, and adding one is borderline scope creep. The doc has a Changelog section though, and adding a dated entry for a "refresh pass" is consistent with the doc's own convention and keeps the "Last updated: 2026-06-22" honest. I'll add one concise entry.

46. user (2026-06-22T09:49:11.592Z)

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47. assistant (2026-06-22T09:49:15.123Z)

All edits applied. Let me do a final scan to confirm no stray "in-flight on branch" or "zero app-layer auth" or "Absent (net-new): GET /api/capabilities" remnants survive elsewhere in the doc.

48. user (2026-06-22T09:49:16.503Z)

14:[Omitted long matching line]
197:- 2026-06-22 ? Freshness pass: `DRAFT`?`IN PROGRESS`; re-anchored engine facts against
`origin/master` ? `/api/capabilities` + `/api/health` now exist (T7 engine half ?);
`native\_wasb\_runner.py` is on master (no longer branch-only); CORS `` is now env-lockable + the
engine ships a Host-guard / optional Cf-Access edge-auth / boot-posture (so "zero app-layer
auth" is no longer literal); alpha deploy went LIVE 2026-06-22 (`app.khelsutra.guru`), ticking
the raw-`:8000`-refused DoD box.

49. assistant (2026-06-22T09:49:19.428Z)

Line 14 matches "zero app-layer auth" ? let me confirm that's my reworded version (which uses
the phrase in the negation "no longer literally true"), not a leftover.

50. user (2026-06-22T09:49:20.184Z)

14 The ****single most important caveat:**** the engine's CORS default is `` but is now ****env-
configurable**** with `allow\_credentials=False` `[verified
backend/main.py:~:cors_origins_from_env, origin/master]`, and it now ships a ****Host-header
guard**** (`TrustedHostMiddleware`) `[verified backend/main.py, origin/master]` + an ****optional
edge-auth integration**** (verifies the `Cf-Access-Jwt-Assertion` JWT) `[verified
backend/api/auth.py, origin/master]` + an ****exposure-safety boot posture****
(`enforce\_startup\_checks`) `[verified backend/config/startup.py, origin/master]` ? so "zero app-
layer auth" is ****no longer literally true****. Even so, safety is still structural ****only****
because the engine stays bound to `127.0.0.1:8000` `[verified backend/main.py:638]`, a firewall
blocks inbound `8000`, and a reverse proxy/tunnel is the sole authenticated ingress ? ****the
gate, not the app, is the tenancy boundary.****

51. assistant (2026-06-22T09:49:24.462Z)

That's my reworded line ? correct and accurate. All five stale findings are surgically addressed
and re-verified against engine `origin/master`. The "unverifiable" perDoc items (T11 retention
#299/#301, droplet filesystem paths, DESKTOP_APP_PLAN supersession) I left untouched since I
could not positively verify them from these repos and the instructions say not to act on
unverifiable items.

52. user (2026-06-22T09:49:58.093Z)

Structured output provided successfully

53. assistant (2026-06-22T09:50:02.816Z)

I've completed the freshness pass on `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md`. All 5 stale
findings were re-verified against the engine's `origin/master` and surgically fixed, plus the
cascading references to each stale fact within the same doc (risk table rows, §4.4 invariant).
One audit detail was slightly off (it cited the wrong Cf-Access header name and a non-existent
`owner\_id` in auth.py) ? I wrote the doc to match the actual master code rather than the audit's
phrasing. The structured report is returned.